

JSON FILE

```
In [90]: import pandas as pd
df = pd.read_json(r"C:\Users\mahim\Downloads\airports.json")
df1 = pd.json_normalize(df["airports"])
```

```
In [91]: print(df1.head())
print(df1.tail())
```

	code	name	city	country	\
0	ATL	Hartsfield-Jackson Atlanta International	Atlanta	US	
1	PEK	Beijing Capital International	Beijing	CN	
2	DXB	Dubai International	Dubai	AE	
3	LAX	Los Angeles International	Los Angeles	US	
4	HND	Tokyo Haneda	Tokyo	JP	

	lat	lon
0	33.6407	-84.4277
1	40.0801	116.5846
2	25.2528	55.3644
3	33.9425	-118.4081
4	35.5494	139.7798

	code	name	city	country	lat	lon
25	MUC	Munich Airport	Munich	DE	48.3537	11.7750
26	BCN	Barcelona-El Prat	Barcelona	ES	41.2974	2.0833
27	YYZ	Toronto Pearson International	Toronto	CA	43.6777	-79.6248
28	MEX	Mexico City International	Mexico City	MX	19.4361	-99.0719
29	NRT	Narita International	Tokyo	JP	35.7720	140.3929

```
In [92]: print(df1.isnull())
print(df1.isnull().sum())
print(df1["name"].isnull())
print(df.dropna(axis=1))
```

	code	name	city	country	lat	lon
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
5	False	False	False	False	False	False
6	False	False	False	False	False	False
7	False	False	False	False	False	False
8	False	False	False	False	False	False
9	False	False	False	False	False	False
10	False	False	False	False	False	False
11	False	False	False	False	False	False
12	False	False	False	False	False	False
13	False	False	False	False	False	False
14	False	False	False	False	False	False
15	False	False	False	False	False	False
16	False	False	False	False	False	False
17	False	False	False	False	False	False
18	False	False	False	False	False	False
19	False	False	False	False	False	False
20	False	False	False	False	False	False
21	False	False	False	False	False	False
22	False	False	False	False	False	False
23	False	False	False	False	False	False
24	False	False	False	False	False	False
25	False	False	False	False	False	False
26	False	False	False	False	False	False
27	False	False	False	False	False	False
28	False	False	False	False	False	False
29	False	False	False	False	False	False

```
code      0
name      0
city      0
country   0
lat       0
lon       0
dtype: int64
```

```
0      False
1      False
2      False
3      False
4      False
5      False
6      False
7      False
8      False
9      False
10     False
11     False
12     False
13     False
14     False
15     False
16     False
17     False
18     False
19     False
20     False
21     False
```

```

22     False
23     False
24     False
25     False
26     False
27     False
28     False
29     False
Name: name, dtype: bool

```

```

                                airports
0  {'code': 'ATL', 'name': 'Hartsfield-Jackson At...
1  {'code': 'PEK', 'name': 'Beijing Capital Inter...
2  {'code': 'DXB', 'name': 'Dubai International',...
3  {'code': 'LAX', 'name': 'Los Angeles Internati...
4  {'code': 'HND', 'name': 'Tokyo Haneda', 'city'...
5  {'code': 'ORD', 'name': 'O'Hare International'...
6  {'code': 'LHR', 'name': 'London Heathrow', 'ci...
7  {'code': 'PVG', 'name': 'Shanghai Pudong Inter...
8  {'code': 'CDG', 'name': 'Paris Charles de Gaul...
9  {'code': 'DFW', 'name': 'Dallas/Fort Worth Int...
10 {'code': 'CAN', 'name': 'Guangzhou Baiyun Inte...
11 {'code': 'AMS', 'name': 'Amsterdam Schiphol', ...
12 {'code': 'FRA', 'name': 'Frankfurt Airport', '...
13 {'code': 'IST', 'name': 'Istanbul Airport', 'c...
14 {'code': 'SIN', 'name': 'Singapore Changi', 'c...
15 {'code': 'ICN', 'name': 'Incheon International...
16 {'code': 'DEN', 'name': 'Denver International'...
17 {'code': 'BKK', 'name': 'Suvarnabhumi Airport'...
18 {'code': 'JFK', 'name': 'John F. Kennedy Inter...
19 {'code': 'HKG', 'name': 'Hong Kong Internation...
20 {'code': 'KUL', 'name': 'Kuala Lumpur Internat...
21 {'code': 'MAD', 'name': 'Adolfo Suárez Madrid-...
22 {'code': 'SFO', 'name': 'San Francisco Interna...
23 {'code': 'MIA', 'name': 'Miami International',...
24 {'code': 'SYD', 'name': 'Sydney Kingsford Smit...
25 {'code': 'MUC', 'name': 'Munich Airport', 'cit...
26 {'code': 'BCN', 'name': 'Barcelona-El Prat', '...
27 {'code': 'YYZ', 'name': 'Toronto Pearson Inter...
28 {'code': 'MEX', 'name': 'Mexico City Internati...
29 {'code': 'NRT', 'name': 'Narita International'...

```

```
In [93]: df1["name"].fillna("Unknown")
```

```

Out[93]: 0      Hartsfield-Jackson Atlanta International
        1      Beijing Capital International
        2      Dubai International
        3      Los Angeles International
        4      Tokyo Haneda
        5      O'Hare International
        6      London Heathrow
        7      Shanghai Pudong International
        8      Paris Charles de Gaulle
        9      Dallas/Fort Worth International
       10      Guangzhou Baiyun International
       11      Amsterdam Schiphol
       12      Frankfurt Airport
       13      Istanbul Airport
       14      Singapore Changi
       15      Incheon International
       16      Denver International
       17      Suvarnabhumi Airport
       18      John F. Kennedy International
       19      Hong Kong International
       20      Kuala Lumpur International
       21      Adolfo Suárez Madrid-Barajas
       22      San Francisco International
       23      Miami International
       24      Sydney Kingsford Smith
       25      Munich Airport
       26      Barcelona-El Prat
       27      Toronto Pearson International
       28      Mexico City International
       29      Narita International
Name: name, dtype: object

```

```

In [94]: print(df1.shape)
         print(df1.info())
         print(df1.describe())

```

```

(30, 6)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   code        30 non-null    object
 1   name        30 non-null    object
 2   city        30 non-null    object
 3   country     30 non-null    object
 4   lat         30 non-null    float64
 5   lon         30 non-null    float64
dtypes: float64(2), object(4)
memory usage: 1.5+ KB
None

```

	lat	lon
count	30.000000	30.000000
mean	31.951923	16.402787
std	17.946009	93.216910
min	-33.939900	-122.379000
25%	25.388575	-80.121450
50%	36.616100	6.665250
75%	41.291875	110.971975
max	52.310500	151.175300

EXCEL FILE

```
In [95]: df = pd.read_excel("C:/Users/mahim/Downloads/Historicalinvesttemp.xlsx")
print(df.head())
print(df.tail())
print(df.info())
print(df.shape)
print(df.describe())
```

```

      Unnamed: 0      Unnamed: 1 Unnamed: 2 Unnamed: 3
0      NaN      NaN      NaN      NaN
1      NaN      NaN      NaN      NaN
2      NaN      NaN      NaN      NaN
3      NaN      NaN      NaN      NaN
4      NaN      Annual Returns on Investments in      NaN      NaN
      Unnamed: 0 Unnamed: 1 Unnamed: 2 Unnamed: 3
85      2007      0.0549      0.0988      0.0466
86      2008      -0.37      0.2587      0.016
87      2009      0.2646      -0.149      0.001
88      NaN      stocks      tbills      bonds
89      averages      NaN      NaN      NaN
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0  84 non-null    object
1   Unnamed: 1  85 non-null    object
2   Unnamed: 2  84 non-null    object
3   Unnamed: 3  84 non-null    object
dtypes: object(4)
memory usage: 2.9+ KB
None
(90, 4)
      Unnamed: 0      Unnamed: 1 Unnamed: 2 Unnamed:
3
count      84      85      84.0000      84.000
0
unique      84      85      80.0000      83.000
0
top      Year      Annual Returns on Investments in      0.0038      0.046
6
freq      1      1      5.0000      2.000
0
```

```
In [96]: print(df.isnull())
print(df.isnull().sum())
```

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3
0	True	True	True	True
1	True	True	True	True
2	True	True	True	True
3	True	True	True	True
4	True	False	True	True
..	...	...	...	...
85	False	False	False	False
86	False	False	False	False
87	False	False	False	False
88	True	False	False	False
89	False	True	True	True

[90 rows x 4 columns]

Unnamed: 0 6

Unnamed: 1 5

Unnamed: 2 6

Unnamed: 3 6

dtype: int64

```
In [97]: import pandas as pd
df=pd.read_csv("C:/Users/mahim/Downloads/eda and ml dataset/store_sales.c
print(df)
```

CustomerID	Age	Gender	Category	ItemPurchased	Amount	Season	\
1	58	Female	Accessories	Handbag	115.50	Autumn	
2	40	Male	Mens Clothing	Shirt	103.43	Spring	
3	66	Female	Sports	Football	35.45	Spring	
4	39	Female	Accessories	Handbag	153.31	Spring	
5	23	Female	Home	Curtains	151.43	Winter	
...	...	...	...	...	...	...	
4996	44	Female	Accessories	Sunglasses	172.64	Winter	
4997	59	Male	Groceries	Beverages	54.00	Spring	
4998	41	Female	Accessories	Handbag	159.33	Autumn	
4999	63	Female	Womens Clothing	Skirt	173.82	Autumn	
5000	41	Female	Footwear	Sandals	55.39	Autumn	

CustomerID	PaymentMethod	ItemRating	DiscountApplied(%)	\
1	Card	3.5	18	
2	Card	4.1	13	
3	Card	3.3	11	
4	Card	4.4	13	
5	Card	4.1	20	
...	...	...	...	
4996	Card	3.6	20	
4997	Cash on Delivery	3.8	14	
4998	Card	4.3	26	
4999	Card	4.8	26	
5000	Cash on Delivery	4.1	19	

CustomerID	PreviousPurchases
1	4
2	4
3	3
4	4
5	10
...	...
4996	5
4997	2
4998	8
4999	8
5000	7

[5000 rows x 10 columns]

accessing

```
In [98]: print(df.loc[333,['Gender','ItemPurchased']])
```

```
Gender      Male
ItemPurchased  Curtains
Name: 333, dtype: object
```

```
In [99]: print(df.loc[333:340,['Gender','ItemPurchased']])
```

CustomerID	Gender	ItemPurchased
333	Male	Curtains
334	Female	Skirt
335	Female	Wallet
336	Female	Sunglasses
337	Female	Top
338	Male	Shirt
339	Female	Jeans
340	Female	Wallet

groupby

```
In [100... print(df.groupby('Gender').agg({'Amount': ['min', 'max', 'mean']}))
```

Gender	Amount		
	min	max	mean
Female	5.89	299.25	110.119269
Male	5.08	2997.94	460.622580

transform

```
In [101... df['a'] = (df['Amount'] - df.groupby('Gender')['Amount'].transform('mean'))
print(df[['Gender', 'Amount', 'a']].head())
```

CustomerID	Gender	Amount	a
1	Female	115.50	5.380731
2	Male	103.43	-357.192580
3	Female	35.45	-74.669269
4	Female	153.31	43.190731
5	Female	151.43	41.310731

interpolate

```
In [102... a=pd.DataFrame({'age': [5,10,None,20,25]})
a['age']=a['age'].interpolate()
print(a)
```

	age
0	5.0
1	10.0
2	15.0
3	20.0
4	25.0

merge

```
In [103... import pandas as pd
data = r"C:\Users\mahim\Downloads\Merge_Join_Concat_Dataset_50_Records.xls"
students = pd.read_excel(data, sheet_name="students")
marks = pd.read_excel(data, sheet_name="marks")
contact = pd.read_excel(data, sheet_name="contact")
print(students.head())
print(marks.head())
print(contact.head())
```


	Student_ID	Name	Department
0	1001	Student_1	CSE
1	1002	Student_2	IT
2	1003	Student_3	ECE
3	1004	Student_4	EEE
4	1005	Student_5	MECH

	Student_ID	Maths	Python
0	1001	61	56
1	1002	62	57
2	1003	63	58
3	1004	64	59
4	1005	65	60

	Student_ID	Email
0	1001	student1@college.edu
1	1002	student2@college.edu
2	1003	student3@college.edu
3	1004	student4@college.edu
4	1005	student5@college.edu

inner join

```
In [104... df=pd.merge(students,marks,on="Student_ID",how="inner")
print(idf.head())
print(idf.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61.0	56.0
1	1002	Student_2	IT	62.0	57.0
2	1003	Student_3	ECE	63.0	58.0
3	1004	Student_4	EEE	64.0	59.0
4	1005	Student_5	MECH	65.0	60.0

(55, 5)

left join

```
In [105... ldf=pd.merge(students,marks,on="Student_ID",how="left")
print(idf.head())
print(idf.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61.0	56.0
1	1002	Student_2	IT	62.0	57.0
2	1003	Student_3	ECE	63.0	58.0
3	1004	Student_4	EEE	64.0	59.0
4	1005	Student_5	MECH	65.0	60.0

(55, 5)

right

```
In [106... idf=pd.merge(students,marks,on="Student_ID",how="right")
print(idf.head())
print(idf.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61	56
1	1002	Student_2	IT	62	57
2	1003	Student_3	ECE	63	58
3	1004	Student_4	EEE	64	59
4	1005	Student_5	MECH	65	60

(50, 5)

outer

```
In [107... idf=pd.merge(students,marks,on="Student_ID",how="outer")
print(idf.head())
print(idf.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61.0	56.0
1	1002	Student_2	IT	62.0	57.0
2	1003	Student_3	ECE	63.0	58.0
3	1004	Student_4	EEE	64.0	59.0
4	1005	Student_5	MECH	65.0	60.0

(55, 5)

add the contact

```
In [108... df1=pd.merge(ldf,contact,on="Student_ID",how="left")
print(df1.head())
```

	Student_ID	Name	Department	Maths	Python	Email
0	1001	Student_1	CSE	61.0	56.0	student1@college.edu
1	1002	Student_2	IT	62.0	57.0	student2@college.edu
2	1003	Student_3	ECE	63.0	58.0	student3@college.edu
3	1004	Student_4	EEE	64.0	59.0	student4@college.edu
4	1005	Student_5	MECH	65.0	60.0	student5@college.edu

row

```
In [109... p1=students.iloc[:25]
p2=students.iloc[25:]
c=pd.concat([p1,p2])
print(c.shape)
print(c.head())
```

(50, 3)

	Student_ID	Name	Department
0	1001	Student_1	CSE
1	1002	Student_2	IT
2	1003	Student_3	ECE
3	1004	Student_4	EEE
4	1005	Student_5	MECH

column wise

```
In [110... s1=students[['Student_ID','Name']]
s2=marks[['Python']]
s3=pd.concat([s1,s2],axis=1)
print(s3.head())
```

	Student_ID	Name	Python
0	1001	Student_1	56
1	1002	Student_2	57
2	1003	Student_3	58
3	1004	Student_4	59
4	1005	Student_5	60

```
In [111... s1=students.set_index("Student_ID")
s2=contact.set_index("Student_ID")
s3=s1.join(s2)
```

```
print(s3.head())
```

	Name	Department	Email
Student_ID			
1001	Student_1	CSE	student1@college.edu
1002	Student_2	IT	student2@college.edu
1003	Student_3	ECE	student3@college.edu
1004	Student_4	EEE	student4@college.edu
1005	Student_5	MECH	student5@college.edu

pivot table

```
In [112... df=pd.read_csv("C:/Users/mahim/Downloads/people-100.csv")
print(df.head())
```

	Index	User Id	First Name	Last Name	Sex	\
0	1	88F7B33d2bcf9f5	Shelby	Terrell	Male	
1	2	f90cD3E76f1A9b9	Phillip	Summers	Female	
2	3	DbeAb8CcdfeFC2c	Kristine	Travis	Male	
3	4	A31Bee3c201ef58	Yesenia	Martinez	Male	
4	5	1bA7A3dc874da3c	Lori	Todd	Male	

	Email	Phone	Date of birth	\
0	elijah57@example.net	001-084-906-7849x73518	1945-10-26	
1	bethany14@example.com	214.112.6044x4913	1910-03-24	
2	bthompson@example.com	277.609.7938	1992-07-02	
3	kaitlinkaiser@example.com	584.094.6111	2017-08-03	
4	buchananmanuel@example.net	689-207-3558x7233	1938-12-01	

	Job Title
0	Games developer
1	Phytotherapist
2	Homeopath
3	Market researcher
4	Veterinary surgeon

Create Age Column

```
In [113... df['Date of birth']=pd.to_datetime(df['Date of birth'])
today=pd.Timestamp.today()
df['Age']=((today-df['Date of birth']).dt.days//365)
print(df[['First Name','Sex','Age']].head())
```

	First Name	Sex	Age
0	Shelby	Male	80
1	Phillip	Female	116
2	Kristine	Male	33
3	Yesenia	Male	8
4	Lori	Male	87

Average age by gender and Pivot Table with aggfunc='mean' and multiple and overall average

```
In [114... pivot1=pd.pivot_table(
    df,
    values='Age',
    index='Sex'
)
print(pivot1)
```

```

pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean'
)
print(pivot2)
pivot3=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc=['mean','max','min']
)
print(pivot2)
pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean',
    margins=True
)
print(pivot2)
pivot3=pd.pivot_table(
    df,
    values='Age',
    index='Job Title',
    columns='Sex',
    aggfunc=['mean','max','min']
)

```

```

                Age
Sex
Female  64.981132
Male    54.957447

```

```

                Age
Sex
Female  64.981132
Male    54.957447

```

```

                Age
Sex
Female  64.981132
Male    54.957447

```

```

                Age
Sex
Female  64.981132
Male    54.957447
All      60.270000

```

Data Cleaning

```

In [127... import pandas as pd

df = pd.read_excel("C:/Users/mahim/Downloads/Data_Cleaning_50Plus_Records
print(df.head())
print(df.columns)

```

	Student ID	Student Name	Department	Age	Salary	Email
0	101	student1	IT	20.0	23075.0	student1@gmail.com
1	102	student2	ECE	18.0	35929.0	student2@gmail.com
2	103	STUDENT3	EEE	22.0	29718.0	student3@gmail.com
3	104	student4	MECH	20.0	21009.0	student4@gmail.com
4	105	Student5	CSE	19.0	47113.0	student5@gmail.com

```
Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary', 'Email'], dtype='object')
```

```
In [128... df.rename(columns={
    'Student ID': 'Student_ID',
    'Student Name ': 'Student_Name',
    ' Department': 'Department'
}, inplace=True)

print(df.columns)
```

```
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')
```

changing the data type

```
In [135... df["Student_ID"] = df["Student_ID"].astype(str)
df["Age"] = df["Age"].fillna(0)
df["Age"] = df["Age"].astype(int)
print(df.dtypes)
```

```
Student_ID      object
Student_Name    object
Department      object
Age             int32
Salary          float64
Email           object
dtype: object
```

checking duplicates

```
In [137... d = df.duplicated().sum()
print(d)
```

0